ROUSHAN KUMAR

Software Developer — Backend & Cloud

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PROFESSIONAL SUMMARY

Backend and Full-Stack Developer who builds web scrapers, APIs, and cloud services in Python and TypeScript. Created and runs **AstreWork** (astrework.app) on Oracle Cloud, where Dockerized Playwright and Puppeteer pipelines harvest and deduplicate 14,000+ high-volume career listings daily using Turso SQLite and Redis. Enjoys solving hard concurrency, caching, and infrastructure problems.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, C#, SQL, Bash
- **Backend & APIs:** FastAPI, Node.js, Express.js, .NET Core Web API, REST APIs, Microservices
- **Frontend & Web:** Next.js 16 (App Router), React 19, Redux, Tailwind CSS, HTML5/CSS3
- **Automation & AI:** Playwright, Puppeteer, Selenium, Tree-Sitter AST, Google GenAI SDK, ChromaDB
- **Databases & Caching:** Turso LibSQL (SQLite WAL), PostgreSQL, MongoDB, Redis 7, Prisma ORM
- **Cloud, Security & Testing:** Oracle Cloud (OCI ARM64), Docker, Caddy, GitHub Actions, Web Crypto API, Vitest, PyTest, xUnit

TECHNICAL PROJECTS

AstreWork — Autonomous Career Engine & Distributed Harvester (Next.js 16, Python, OCI, Docker, Redis) **Live Demo | GitHub**

- **Creator & Lead Backend Developer:** Built and run distributed scrapers on Oracle Cloud (OCI ARM64) that harvest and deduplicate 14,000+ career listings daily across 7 platforms.
- **Cloud Infrastructure:** Deployed multi-stage Docker containers on OCI Ampere ARM64 behind Caddy HTTP/3 and Watchtower CI/CD, holding 99.9% uptime and sub-50ms API latency.
- **Storage & Caching:** Designed the data layer across Turso Cloud LibSQL (SQLite WAL) and Redis 7 LRU cache, cutting repeat query latency by 85%.

Astra Vision — AI Code Sandbox & AST Dependency Engine (FastAPI, Python, Tree-Sitter, ChromaDB) **Live Demo | GitHub**

- **Backend Service:** Built a FastAPI code-analysis backend that parses multi-file repositories into syntax trees and call graphs before running untrusted code.
- **Sandbox Isolation:** Wrote a sandboxed Python subprocess runner with syscall and socket guards to intercept unauthorized file and network access and safely terminate rogue processes.
- **AST & Vector Indexing:** Wired Tree-Sitter AST parsers into ChromaDB vector search to resolve cross-file symbol references in under 200ms.

Alyra Lock — Zero-Knowledge Encrypted Password Vault (TypeScript, Express, MongoDB, Web Crypto) **Live Demo | GitHub**

- **Zero-Knowledge API:** Built an Express.js and MongoDB sync backend that stores only ciphertext blobs and IVs so plaintext secrets never leave the client.
- **Cryptographic Core:** Implemented AES-GCM (256-bit) encryption and 100,000-iteration PBKDF2-SHA256 key derivation using the browser Web Crypto API.
- **Auth & Sync Hardening:** Added JWT session rotation, strict CORS rules, and atomic document versioning that completes encrypted syncs in under 200ms.

TECHNICAL SIMULATIONS & VIRTUAL EXPERIENCES

Commonwealth Bank (CommBank) — Software Development Virtual Simulation **June 2026**

- **Problem & Scope:** Diagnosed and resolved silent data-overwrite bugs across high-traffic C#/.NET Core financial API controllers backed by MongoDB.
- **Action Taken:** Refactored C# Web API endpoints with $set atomic operators for partial document updates and patched concurrency race conditions.
- **Engineering Outcome:** Authored backend unit test suites using xUnit and Moq covering boundary conditions to guarantee transactional consistency.

Y Combinator (YC Startup - Shiptivity) — Software Development Virtual Simulation **June 2026**

- **Problem & Scope:** Fixed priority-key collisions and transaction rollbacks during high-frequency task reordering in an embedded SQLite database.
- **Action Taken:** Architected SQLite sequential priority re-indexing using atomic transaction blocks and patched OpenSSL compilation bottlenecks.
- **Engineering Outcome:** Reduced database write lock contention by 40% and improved Node.js build pipeline stability across legacy environments.

Walmart USA — Advanced Software Development Virtual Simulation **May 2026**

- **Problem & Scope:** Solved indexing overhead and latency spikes in high-volume retail inventory processing queues fed by disparate data sources.
- **Action Taken:** Implemented a generic K-ary Max Heap in Java using bitwise shift operators (<<, >>>) for rapid O(log n) priority retrieval.
- **Engineering Outcome:** Accelerated priority queue indexing by 35% and built automated Python ETL scripts (csv, sqlite3) for batch database loading.

EDUCATION

Bachelor of Computer Applications (BCA) — Amity University Noida **2025 – 2028**

- **Relevant Coursework:** Data Structures & Algorithms (DSA) • Database Management Systems (DBMS) • Operating Systems (OS) • Computer Networks • Object-Oriented Programming (OOP)